

Coding Set 2

Problem 1: Student Details

Create a class Student with:

- Private data members: name, rollNo
- Public functions:
 - setData()
 - displayData()

Create an object and display the student's details.

Concepts: Class, Object, Private/Public Access Specifiers

Problem 2: Rectangle Area

Create a class Rectangle with:

- Private data members: length, breadth
- Public functions:
 - input()
 - calculateArea()
 - displayArea()

Calculate and display the area of the rectangle.

Concepts: Class, Object, Functions

Problem 3: Number Analyzer

Create a class Number.

- Private data member: num
- Public functions:
 - input()
 - isEven()
 - displayResult()

Determine whether the number is even or odd.

Concepts: Class, Conditional Statements

Problem 4: Constructor Demonstration

Create a class Book.

Data members:

- title
- author

Use a **parameterized constructor** to initialize both values.

Display the details of the book.

Concepts: Constructors

Problem 5: Sum of Array Elements

Create a class ArraySum.

- Private:
 - integer array of size 10
- Public:
 - constructor to input array elements
 - function findSum()

Display the sum of all elements.

Concepts: Constructor, Arrays, Loops

Problem 6: String Length Finder

Create a class Text.

- Private data member: str
- Public:
 - constructor to accept a string
 - function to calculate string length without using length() or size()

Display the length.

Concepts: Strings, Loops, Constructor

Problem 7: Employee Salary Calculator

Create a class Employee.

Private members:

- employeeName

- basicSalary

Public members:

- constructor
- function to calculate HRA = 20% of salary
- function to calculate DA = 10% of salary
- function to display gross salary

Gross Salary = Basic + HRA + DA

Concepts: Constructor, Arithmetic Operations

Problem 8: Highest Marks in an Array

Create a class Marks.

Private:

- array of marks of 5 students

Public:

- constructor to input marks
- function to find highest mark
- function to display result

Concepts: Arrays, Loops, Constructor

Problem 9: Bank Account

Create a class BankAccount.

Private members:

- accountNumber
- balance

Public members:

- constructor
- deposit(amount)
- withdraw(amount)
- displayBalance()

The withdrawal should occur only if sufficient balance exists.

Concepts: Encapsulation, Access Specifiers, Constructors

Problem 10: Student Result Management System

Create a class Student.

Private members:

- name
- rollNo
- marks[5]

Public members:

- constructor to input details
- function to calculate total marks
- function to calculate percentage
- function to determine grade

Grade Criteria:

Percentage	Grade
90 and above	A
75-89	B
60-74	C
40-59	D
Below 40	F

Display complete result.

Concepts: Classes, Objects, Constructors, Arrays, Loops, Decision Making

Challenge Problem (Bonus)

Create a class Library.

- Store details of 10 books using an array of objects.
- Use constructors to initialize data.
- Search a book by title.
- Display all matching details.

Concepts Covered: Classes, Objects, Constructors, Arrays of Objects, Strings, Loops, Searching.